

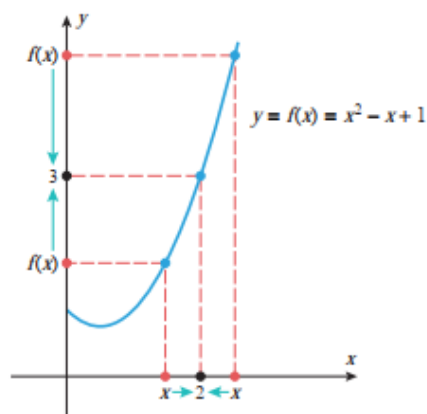
Limit of a function

The most basic use of limits is to describe how a function behaves as the independent variable approaches a given value. For example, let us examine the behavior of the function

$$f(x) = x^2 - x + 1$$

for x -values closer and closer to 2. It is evident from the graph and table in Figure 1.1.8 that the values of $f(x)$ get closer and closer to 3 as values of x are selected closer and closer to 2 on either the left or the right side of 2. We describe this by saying that the “limit of $x^2 - x + 1$ is 3 as x approaches 2 from either side,” and we write

$$\lim_{x \rightarrow 2} (x^2 - x + 1) = 3 \quad (5)$$



x	1.0	1.5	1.9	1.95	1.99	1.995	1.999	2	2.001	2.005	2.01	2.05	2.1	2.5	3.0
$f(x)$	1.000000	1.750000	2.710000	2.852500	2.970100	2.985025	2.997001		3.003001	3.015025	3.030100	3.152500	3.310000	4.750000	7.000000

Left side
Right side

► **Example 2** Use numerical evidence to make a conjecture about the value of

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} \quad (8)$$

Solution. Although the function

$$f(x) = \frac{x - 1}{\sqrt{x} - 1} \quad (9)$$

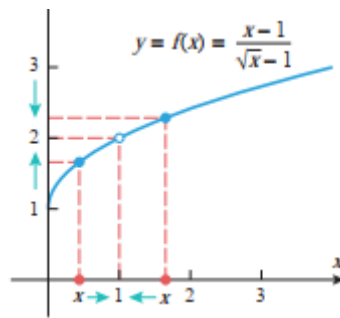
is undefined at $x = 1$, this has no bearing on the limit. Table 1.1.1 shows sample x -values approaching 1 from the left side and from the right side. In both cases the corresponding values of $f(x)$, calculated to six decimal places, appear to get closer and closer to 2, and hence we conjecture that

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} = 2$$

Table 1.1.1

x	0.99	0.999	0.9999	0.99999		1.00001	1.0001	1.001	1.01
$f(x)$	1.994987	1.999500	1.999950	1.999995		2.000005	2.000050	2.000500	2.004988

Left side
Right side



► **Example 3** Use numerical evidence to make a conjecture about the value of

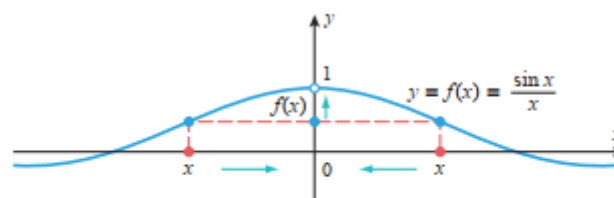
$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \quad (10)$$

Solution. With the help of a calculating utility set in radian mode, we obtain Table 1.1.2. The data in the table suggest that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (11)$$

Table 1.1.2

x (RADIANs)	$y = \frac{\sin x}{x}$
± 1.0	0.84147
± 0.9	0.87036
± 0.8	0.89670
± 0.7	0.92031
± 0.6	0.94107
± 0.5	0.95885
± 0.4	0.97355
± 0.3	0.98507
± 0.2	0.99335
± 0.1	0.99833
± 0.01	0.99998



As x approaches 0 from the left or right, $f(x)$ approaches 1.

EXAMPLE 2 Estimate the value of $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$.

SOLUTION The table lists values of the function for several values of t near 0.

t	$\frac{\sqrt{t^2 + 9} - 3}{t^2}$
± 1.0	0.162277 ...
± 0.5	0.165525 ...
± 0.1	0.166620 ...
± 0.05	0.166655 ...
± 0.01	0.166666 ...

As t approaches 0, the values of the function seem to approach 0.166666 ... and so we guess that

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = \frac{1}{6}$$

EXAMPLE 7 The graph of a function g is shown in Figure 10. Use it to state the values (if they exist) of the following:

- (a) $\lim_{x \rightarrow 2^-} g(x)$ (b) $\lim_{x \rightarrow 2^+} g(x)$ (c) $\lim_{x \rightarrow 2} g(x)$
 (d) $\lim_{x \rightarrow 5^-} g(x)$ (e) $\lim_{x \rightarrow 5^+} g(x)$ (f) $\lim_{x \rightarrow 5} g(x)$

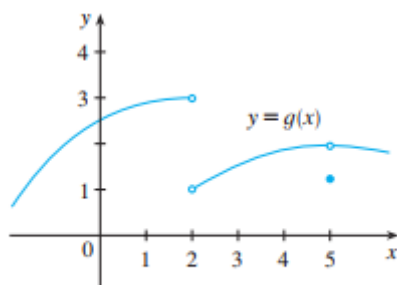


FIGURE 10

SOLUTION From the graph we see that the values of $g(x)$ approach 3 as x approaches 2 from the left, but they approach 1 as x approaches 2 from the right. Therefore

$$(a) \lim_{x \rightarrow 2^-} g(x) = 3 \quad \text{and} \quad (b) \lim_{x \rightarrow 2^+} g(x) = 1$$

(c) Since the left and right limits are different, we conclude from (3) that $\lim_{x \rightarrow 2} g(x)$ does not exist.

The graph also shows that

$$(d) \lim_{x \rightarrow 5^-} g(x) = 2 \quad \text{and} \quad (e) \lim_{x \rightarrow 5^+} g(x) = 2$$

(f) This time the left and right limits are the same and so, by (3), we have

$$\lim_{x \rightarrow 5} g(x) = 2$$

Despite this fact, notice that $g(5) \neq 2$.

EXAMPLE 3 Estimating a Limit Numerically

Use a table to estimate the limit numerically.

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1}$$

Solution Let $f(x) = x/(\sqrt{x+1} - 1)$. Then construct a table that shows values of $f(x)$ for two sets of x -values—one that approaches 0 from the left and one that approaches 0 from the right.

x	-0.01	-0.001	-0.0001	0	0.0001	0.001	0.01
$f(x)$	1.99499	1.99950	1.99995	?	2.00005	2.00050	2.00499

From the table, it appears that the limit is 2. Figure 12.2 verifies this conclusion.

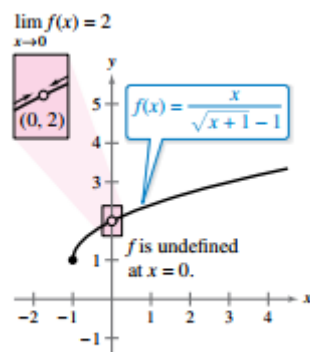


Figure 12.2

► **Example 11** Let

$$f(x) = \begin{cases} 1/(x+2), & x < -2 \\ x^2 - 5, & -2 < x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$$

Find

$$(a) \lim_{x \rightarrow -2} f(x) \quad (b) \lim_{x \rightarrow 0} f(x) \quad (c) \lim_{x \rightarrow 3} f(x)$$

Solution (a). We will determine the stated two-sided limit by first considering the corresponding one-sided limits. For each one-sided limit, we must use that part of the formula that is applicable on the interval over which x varies. For example, as x approaches -2 from the left, the applicable part of the formula is

$$f(x) = \frac{1}{x+2}$$

and as x approaches -2 from the right, the applicable part of the formula near -2 is

$$f(x) = x^2 - 5$$

Thus,

$$\begin{aligned} \lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} \frac{1}{x+2} = -\infty \\ \lim_{x \rightarrow -2^+} f(x) &= \lim_{x \rightarrow -2^+} (x^2 - 5) = (-2)^2 - 5 = -1 \end{aligned}$$

from which it follows that $\lim_{x \rightarrow -2} f(x)$ does not exist.

Solution (b). The applicable part of the formula is $f(x) = x^2 - 5$ on both sides of 0, so there is no need to consider one-sided limits here. We see directly that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x^2 - 5) = 0^2 - 5 = -5$$

Solution (c). Using the applicable parts of the formula for $f(x)$, we obtain

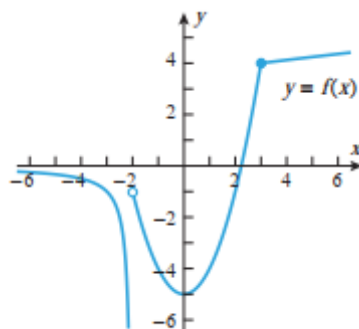
$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 - 5) = 3^2 - 5 = 4$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \sqrt{x + 13} = \sqrt{\lim_{x \rightarrow 3^+} (x + 13)} = \sqrt{3 + 13} = 4$$

Since the one-sided limits are equal, we have

$$\lim_{x \rightarrow 3} f(x) = 4$$

We note that the limit calculations in parts (a), (b), and (c) are consistent with the graph of f shown in Figure 1.2.5. ◀



▲ Figure 1.2.5

EXAMPLE 6 Comparing Left and Right Behavior

Show that the limit does not exist.

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

Solution Consider the graph of $f(x) = |x|/x$, shown in Figure 12.4. Notice that for positive x -values

$$\frac{|x|}{x} = 1, \quad x > 0$$

and for negative x -values

$$\frac{|x|}{x} = -1, \quad x < 0.$$

This means that no matter how close x gets to 0, there are both positive and negative x -values that yield $f(x) = 1$ and $f(x) = -1$, respectively. This implies that the limit does not exist.

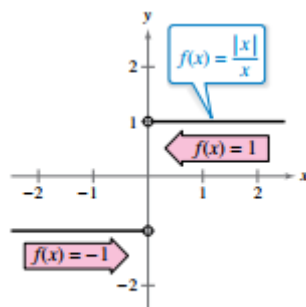


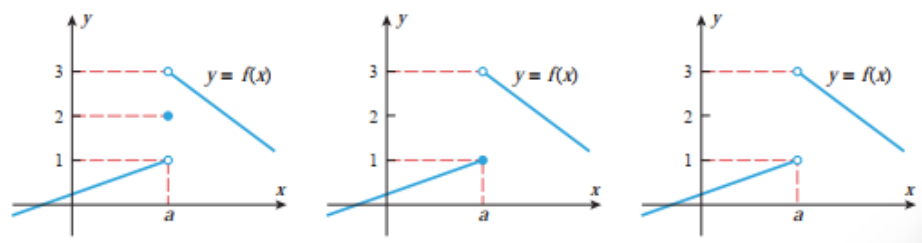
Figure 12.4

► **Example 5** For the functions in Figure 1.1.13, find the one-sided and two-sided limits at $x = a$ if they exist.

Solution. The functions in all three figures have the same one-sided limits as $x \rightarrow a$, since the functions are identical, except at $x = a$. These limits are

$$\lim_{x \rightarrow a^+} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x) = 1$$

In all three cases the two-sided limit does not exist as $x \rightarrow a$ because the one-sided limits are not equal. ◀



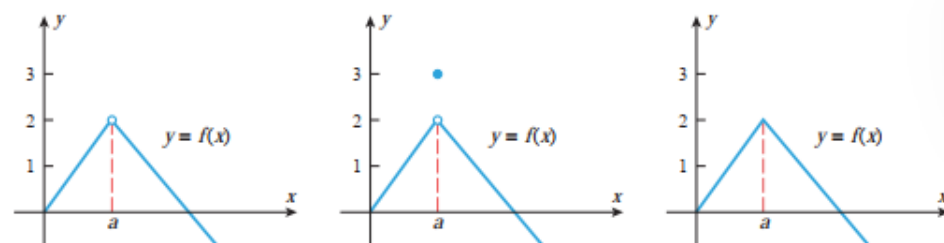
► **Example 6** For the functions in Figure 1.1.14, find the one-sided and two-sided limits at $x = a$ if they exist.

Solution. As in the preceding example, the value of f at $x = a$ has no bearing on the limits as $x \rightarrow a$, so in all three cases we have

$$\lim_{x \rightarrow a^+} f(x) = 2 \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x) = 2$$

Since the one-sided limits are equal, the two-sided limit exists and

$$\lim_{x \rightarrow a} f(x) = 2 \quad \blacktriangleleft$$



EXAMPLE 7 Evaluating One-Sided Limits

Find the limit of $f(x)$ as x approaches 1.

$$f(x) = \begin{cases} 4 - x, & x < 1 \\ 4x - x^2, & x > 1 \end{cases}$$

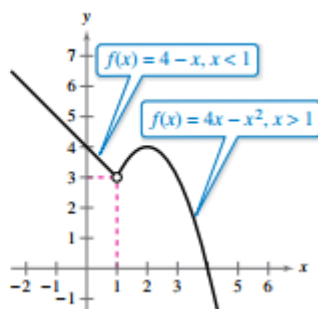
Solution Remember that you are concerned about the value of f near $x = 1$ rather than at $x = 1$. So, for $x < 1$, $f(x)$ is given by $4 - x$. Use direct substitution to obtain

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (4 - x) \\ &= 4 - 1 \\ &= 3. \end{aligned}$$

For $x > 1$, $f(x)$ is given by $4x - x^2$. Use direct substitution to obtain

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (4x - x^2) \\ &= 4(1) - 1^2 \\ &= 3. \end{aligned}$$

Both one-sided limits exist and are equal to 3, so it follows that $\lim_{x \rightarrow 1} f(x) = 3$. The graph at the right confirms this conclusion.



Practice Questions

Q. Complete the table and make a guess about the limit indicated.

11. $f(x) = \frac{e^x - 1}{x}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.01	-0.001	-0.0001	0.0001	0.001	0.01
$f(x)$						

12. $f(x) = \frac{\sin^{-1} 2x}{x}$; $\lim_{x \rightarrow 0} f(x)$

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$						

Q. Find the following using one-sided limits:

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3}$ (b) $\lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12}$ (c) $\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$

31. Let

$$f(x) = \begin{cases} x - 1, & x \leq 3 \\ 3x - 7, & x > 3 \end{cases}$$

Find

(a) $\lim_{x \rightarrow 3^-} f(x)$ (b) $\lim_{x \rightarrow 3^+} f(x)$ (c) $\lim_{x \rightarrow 3} f(x)$.

Use a table to estimate the limit numerically.

$$\lim_{x \rightarrow 1} \frac{x - 1}{x^2 + 3x - 4}$$



Show that the limit does not exist.

$$\lim_{x \rightarrow 1} \frac{2|x - 1|}{x - 1}$$



Estimating a Limit Numerically In Exercises 5–10, complete the table and use the result to estimate the limit numerically. Determine whether it is possible to reach the limit.

5. $\lim_{x \rightarrow 1} (7x + 3)$

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$				?			

6. $\lim_{x \rightarrow -1} (3x^2 + 2x - 6)$

x	-1.1	-1.01	-1.001	-1
$f(x)$				?

x	-0.999	-0.99	-0.9
$f(x)$			

7. $\lim_{x \rightarrow -2} \frac{x+2}{x^2-4}$

x	-2.1	-2.01	-2.001	-2
$f(x)$				?

x	-1.999	-1.99	-1.9
$f(x)$			

8. $\lim_{x \rightarrow 3} \frac{x-3}{x^2-2x-3}$

x	2.9	2.99	2.999	3	3.001	3.01	3.1
$f(x)$				?			

9. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$

x	-0.1	-0.01	-0.001	0
$f(x)$				?

x	0.001	0.01	0.1
$f(x)$			

10. $\lim_{x \rightarrow 0} \frac{\tan x}{2x}$

x	-0.1	-0.01	-0.001	0
$f(x)$				?

x	0.001	0.01	0.1
$f(x)$			

Find the limit of $f(x)$ as x approaches -1 .

$$f(x) = \begin{cases} -x^2 - 3x, & x < -1 \\ x + 3, & x > -1 \end{cases}$$



Evaluating One-Sided Limits In Exercises 35–38, graph the function. Find the limit (if it exists) by evaluating the corresponding one-sided limits.

35. $\lim_{x \rightarrow 1} \frac{|x-1|}{x-1}$

36. $\lim_{x \rightarrow -3} \frac{|x+3|}{x}$

37. $\lim_{x \rightarrow 2} f(x)$, where $f(x) = \begin{cases} 3x-2, & x < 2 \\ 8-x^2, & x \geq 2 \end{cases}$

38. $\lim_{x \rightarrow 3} f(x)$, where $f(x) = \begin{cases} x^2-4, & x \leq 3 \\ x+3, & x > 3 \end{cases}$

Continuity of a Function

Intuitively, the graph of a function can be described as a “continuous curve” if it has no breaks or holes. To make this idea more precise we need to understand what properties of a function can cause breaks or holes. Referring to Figure 1.5.1, we see that the graph of a function has a break or hole if any of the following conditions occur:

1.5.1 DEFINITION A function f is said to be **continuous at $x = c$** provided the following conditions are satisfied:

1. $f(c)$ is defined.
2. $\lim_{x \rightarrow c} f(x)$ exists.
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

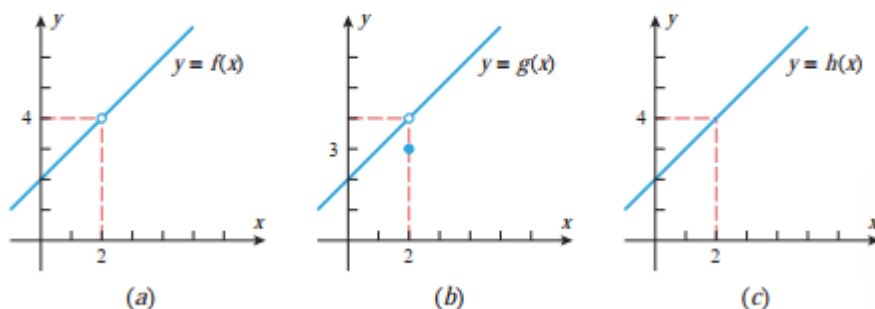
► **Example 1** Determine whether the following functions are continuous at $x = 2$.

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad g(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 3, & x = 2, \end{cases} \quad h(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$$

Solution. In each case we must determine whether the limit of the function as $x \rightarrow 2$ is the same as the value of the function at $x = 2$. In all three cases the functions are identical, except at $x = 2$, and hence all three have the same limit at $x = 2$, namely,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

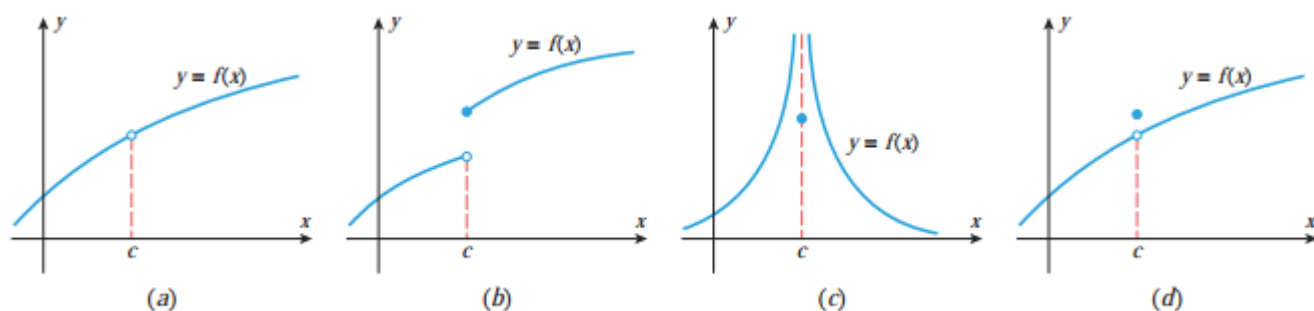
The function f is undefined at $x = 2$, and hence is not continuous at $x = 2$ (Figure 1.5.2a). The function g is defined at $x = 2$, but its value there is $g(2) = 3$, which is not the same as the limit as x approaches 2; hence, g is also not continuous at $x = 2$ (Figure 1.5.2b). The value of the function h at $x = 2$ is $h(2) = 4$, which is the same as the limit as x approaches 2; hence, h is continuous at $x = 2$ (Figure 1.5.2c). (Note that the function h could have been written more simply as $h(x) = x + 2$, but we wrote it in piecewise form to emphasize its relationship to f and g .) ◀



▲ Figure 1.5.2

If one or more of the conditions of this definition fails to hold, then we will say that f has a **discontinuity at $x = c$** . Each function drawn in Figure 1.5.1 illustrates a discontinuity at $x = c$. In Figure 1.5.1a, the function is not defined at c , violating the first condition of Definition 1.5.1. In Figure 1.5.1b, the one-sided limits of $f(x)$ as x approaches c both exist but are not equal. Thus, $\lim_{x \rightarrow c} f(x)$ does not exist, and this violates the second condition of Definition 1.5.1. We will say that a function like that in Figure 1.5.1b has a **jump discontinuity** at c . In Figure 1.5.1c, the one-sided limits of $f(x)$ as x approaches c are infinite. Thus, $\lim_{x \rightarrow c} f(x)$ does not exist, and this violates the second condition of Definition 1.5.1. We will say that a function like that in Figure 1.5.1c has an **infinite discontinuity** at c . In Figure 1.5.1d, the function is defined at c and $\lim_{x \rightarrow c} f(x)$ exists, but these two values are not equal, violating the third condition of Definition 1.5.1. We will

say that a function like that in Figure 1.5.1d has a **removable discontinuity** at c . Exercises 33 and 34 help to explain why discontinuities of this type are given this name.



▲ Figure 1.5.1

► **Example 2** What can you say about the continuity of the function $f(x) = \sqrt{9 - x^2}$?

Solution. Because the natural domain of this function is the closed interval $[-3, 3]$, we will need to investigate the continuity of f on the open interval $(-3, 3)$ and at the two endpoints. If c is any point in the interval $(-3, 3)$, then it follows from Theorem 1.2.2(e) that

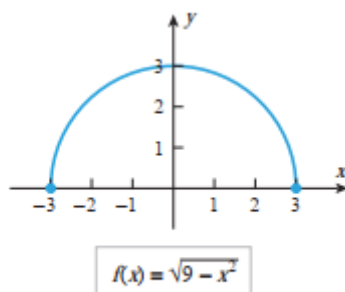
$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow c} (9 - x^2)} = \sqrt{9 - c^2} = f(c)$$

which proves f is continuous at each point in the interval $(-3, 3)$. The function f is also continuous at the endpoints since

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow 3^-} (9 - x^2)} = 0 = f(3)$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9 - x^2} = \sqrt{\lim_{x \rightarrow -3^+} (9 - x^2)} = 0 = f(-3)$$

Thus, f is continuous on the closed interval $[-3, 3]$ (Figure 1.5.5). ◀



▲ Figure 1.5.5

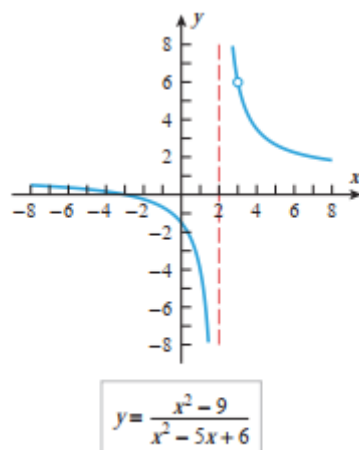
► **Example 3** For what values of x is there a discontinuity in the graph of

$$y = \frac{x^2 - 9}{x^2 - 5x + 6}?$$

Solution. The function being graphed is a rational function, and hence is continuous at every number where the denominator is nonzero. Solving the equation

$$x^2 - 5x + 6 = 0$$

yields discontinuities at $x = 2$ and at $x = 3$ (Figure 1.5.6). ◀



▲ Figure 1.5.6

EXAMPLE 2 Where are each of the following functions discontinuous?

$$\begin{aligned}
 \text{(a) } f(x) &= \frac{x^2 - x - 2}{x - 2} & \text{(b) } f(x) &= \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \\
 \text{(c) } f(x) &= \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases} & \text{(d) } f(x) &= \llbracket x \rrbracket
 \end{aligned}$$

SOLUTION

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2. Later we'll see why f is continuous at all other numbers.

(b) Here $f(0) = 1$ is defined but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2}$$

does not exist. (See Example 1.5.8.) So f is discontinuous at 0.

(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

exists. But

$$\lim_{x \rightarrow 2} f(x) \neq f(2)$$

so f is not continuous at 2.

(d) The greatest integer function $f(x) = \llbracket x \rrbracket$ has discontinuities at all of the integers because $\lim_{x \rightarrow n} \llbracket x \rrbracket$ does not exist if n is an integer. (See Example 1.6.10 and Exercise 1.6.53.)



EXAMPLE 4 Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

SOLUTION If $-1 < a < 1$, then using the Limit Laws, we have

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) \\&= 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} \quad (\text{by Laws 2 and 7}) \\&= 1 - \sqrt{\lim_{x \rightarrow a} (1 - x^2)} \quad (\text{by 11}) \\&= 1 - \sqrt{1 - a^2} \quad (\text{by 2, 7, and 9}) \\&= f(a)\end{aligned}$$

Thus, by Definition 1, f is continuous at a if $-1 < a < 1$. Similar calculations show that

$$\lim_{x \rightarrow -1^+} f(x) = 1 = f(-1) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = 1 = f(1)$$

so f is continuous from the right at -1 and continuous from the left at 1 . Therefore, according to Definition 3, f is continuous on $[-1, 1]$.

The graph of f is sketched in Figure 4. It is the lower half of the circle

$$x^2 + (y - 1)^2 = 1$$

Practice Questions

5. Consider the functions

$$f(x) = \begin{cases} 1, & x \neq 4 \\ -1, & x = 4 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 4x - 10, & x \neq 4 \\ -6, & x = 4 \end{cases}$$

In each part, is the given function continuous at $x = 4$?

- (a) $f(x)$ (b) $g(x)$ (c) $-g(x)$ (d) $|f(x)|$
(e) $f(x)g(x)$ (f) $g(f(x))$ (g) $g(x) - 6f(x)$

6. Consider the functions

$$f(x) = \begin{cases} 1, & 0 \leq x \\ 0, & x < 0 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 0, & 0 \leq x \\ 1, & x < 0 \end{cases}$$

In each part, is the given function continuous at $x = 0$?

- (a) $f(x)$ (b) $g(x)$ (c) $f(-x)$ (d) $|g(x)|$
(e) $f(x)g(x)$ (f) $g(f(x))$ (g) $f(x) + g(x)$

11–22 Find values of x , if any, at which f is not continuous.

11. $f(x) = 5x^4 - 3x + 7$ 12. $f(x) = \sqrt[3]{x-8}$

13. $f(x) = \frac{x+2}{x^2+4}$ 14. $f(x) = \frac{x+2}{x^2-4}$

15. $f(x) = \frac{x}{2x^2+x}$ 16. $f(x) = \frac{2x+1}{4x^2+4x+5}$

17. $f(x) = \frac{3}{x} + \frac{x-1}{x^2-1}$ 18. $f(x) = \frac{5}{x} + \frac{2x}{x+4}$

19. $f(x) = \frac{x^2+6x+9}{|x|+3}$ 20. $f(x) = \left| 4 - \frac{8}{x^4+x} \right|$

21. $f(x) = \begin{cases} 2x+3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$

22. $f(x) = \begin{cases} \frac{3}{x-1}, & x \neq 1 \\ 3, & x = 1 \end{cases}$